# Machine Learning Mini Project Synopsis

## Explainable Multi-Class Network Intrusion Detection System Using Machine Learning


**Group Details**

**Panel C**

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## 1. Abstract

Demand for the network infrastructure, along with IoT, has been growing rapidly. It provides cybercriminals with more chances to launch their attacks. Thus, the evolution of the threat landscape has led to the emergence of new attacks, such as Denial of Service attacks, remote-to-local exploitation, user-to-root privilege escalation, and probing. Traditional intrusion detection systems work with predefined attack patterns only and, therefore, can recognize already known attacks but are not able to recognize any modified ones.

This project proposes an Explainable Multi-Class Network Intrusion Detection System (NIDS) based on supervised machine learning principles. One of the major characteristics of the research is a detailed attack classification instead of the normal vs attack system. For this model, the NSL-KDD dataset will be used along with several measures of effectiveness, including accuracy, precision, recall, and F1-score.

The proposed system operates in restricted spaces; a technique of dimensionality reduction through feature selection is used. It reduces the computational requirements. The workflow uses techniques of interpretation, like feature importance, which helps understand how the model makes decisions, which in turn helps analysts to trust the results during system operations.



**Keywords:** Network Intrusion Detection System, Explainable Artificial Intelligence, NSL-KDD Dataset, Multiclass Classification

## 2. Introduction

Networked infrastructures have become foundational elements of today's organizational processes, covering cloud computing, banking, and massive data transfers. With ever-growing reliance on digital networks, cybersecurity issues have grown extremely uncertain. Cyber-attack threats like DoS, probing, R2L attacks, and U2R privilege escalation are different categories of risks that could result in system disturbances and leakage of sensitive data.

In order to ensure protection against these risks, organizations use IDS solutions, which monitor potential threats like forms of abnormal network activity. Traditional IDS systems are built using catalogued attacks with patterns stored within specific databases, which makes them efficient against known attacks; however, their main drawback lies in the inability to identify novel or modified attacks.

Machine learning models improve threat detection by focusing on not only recognizing patterns and detecting past attacks, but also identifying new ones based on learned behavior. ML models work with historical data, which allows them to detect all types of threats, and multi-class classifiers help identify the exact type of attack, which assists us in further analysis.

The current research is dedicated to the implementation of an Explainable Multi-Class Network IDS on the basis of supervised ML methods. Our main goal is achieving high accuracy in detecting attacks while also giving insights into threat classification.

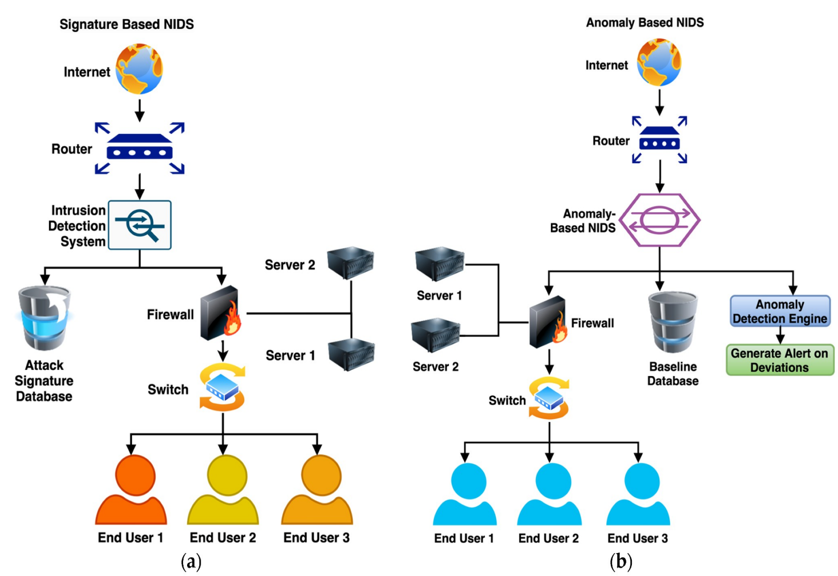


Figure 1: The Improved Network Intrusion Detection Techniques Using the Feature Engineering Approach with Boosting Classifiers

## 3. Literature Review Table

Many research papers are increasingly prioritizing the application of machine learning and deep learning paradigms to Intrusion Detection Systems (IDS). Researchers have scrutinized diverse methodologies, where each methodology offers distinct architectural advantages. They also simultaneously address inherent technical challenges. The following literature review table describes several recent research papers, synthesizing their specific methods used in studies alongside their respective merits and operational limitations.

| Sr. No. | Author(s) & Year | Title / Paper Focus | Method Used | Advantages | Disadvantages |
|---|---|---|---|---|---|
| 1 | **Wang et al. (2020)** | An Explainable Machine Learning Framework for Intrusion Detection Systems | Approach to Explainable ML: SHAP | Intrusion detection systems become more transparent and trustworthy | Emphasis on explainability is greater than that on lightweight implementation. |
| 2 | **Shtayat et al. (2023)** | An Explainable Ensemble Deep Learning Approach for Intrusion Detection in IIoT | SHAP and LIME combined with ensemble deep learning | Robustness and interpretability improved significantly. | Higher complexity that makes practical implementation difficult. |
| 3 | **Sadhwani et al. (2025)** | IoT-based Intrusion Detection System Using Explainable Multi-Class Deep Learning Approaches | CNN, LSTM, Bi-LSTM with SHAP | High multiclass classification accuracy achieved with reduced feature set | Complexity of deep models could require more computational power. |
| 4 | **AlFuraih et al. (2026)** | Explainable Hybrid CNN–XGBoost Framework for Multi-Class IoT Intrusion Detection | CNN, XGBoost with SHAP | Excellent hybrid performance and effective feature selection. | Explanations are not direct and occur in latent space. |

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| 5 | **Sharma et al. (2024)** | Explainable Artificial Intelligence for Intrusion Detection in IoT Networks | DNN, CNN with LIME and SHAP | Good explainability and feature reduction. | Lack of generalization for various attack types. |
| 6 | **Islam et al. (2026)** | An Ensemble Learning Approach for Multi-Class Intrusion Detection with Explainability Analysis | RF, XGBoost, LightGBM, CatBoost with LIME | Very high accuracy and few false positives | The algorithm might work well only on a particular data set. |
| 7 | **Molla et al. (2025)** | Explainable AI Framework for Multiclass Attack Classification in Wireless NIDS | RF, XGBoost, DT, and KNN, with SHAP and LIME | High accuracy and interpretability are provided. | There is Limited evaluation on broader real-world traffic |

Recent research shows that intrusion detection is progressing towards advanced approaches that incorporate multi-class classification, hybrid deep learning models, ensemble techniques, and integrating explainable AI. SHAP and LIME are now being used widely to explain the decisions of AI models. Nevertheless, certain challenges still remain, such as building lightweight models for the real-world, ensuring performance across different environments, handling imbalanced datasets, and balancing model complexity with efficiency.

# 4. Research Gap

Despite significant progress, the implementation of modern intrusion detection systems are still limited by various factors:

- The need for considerable computational resources which is difficult to deploy in resource-limited environments.
- Interpretability is frequently added later rather than being a fundamental design component. This limits its effectiveness in real-time security analysis.
- Many frameworks lack versatility as they are designed for specific areas like wireless or industrial IoT rather than supporting broad-spectrum detection.
- Detecting minority attacks like U2R and R2L remain challenging due to the strong class imbalance present in many training datasets.
- The complexity of ensemble and deep learning models often makes them tough to replicate or even modify, particularly in resource-constrained or academic environments.

## 4.1 Deficiencies in Existing Frameworks

Current industry solutions consistently lack a cohesive integration of:

### 4.1.1 What is Missing in Current Solutions

What is still missing in many current solutions is:

- an accurate system
- **multi-class**
- **explainable**
- **efficient**
- and being **practical enough** for realistic deployment

## 4.2 How Our Proposed Work Addresses It

Our proposed system aims to fill this gap by developing an Explainable Multi-Class Network Intrusion Detection System using Machine Learning that:

- Classifies traffic into **multiple attack categories**
- Uses **feature selection** to reduce complexity
- Integrates **explainability techniques** such as SHAP and feature importance
- Ensures a good balance between **accuracy, interpretability, and efficiency**
- Can be adapted for **lightweight and real-time network monitoring**

## 5. Objectives

The main objectives of the proposed research are:

- To develop a Machine Learning-based Intrusion Detection System for identifying malicious traffic in networks.
- To perform multi-class classification of network attacks instead of binary attack detection.
- To identify major attack categories such as **DoS, Probe, R2L, U2R**, or equivalent network attack classes, based on the dataset used.
- To apply feature selection and feature importance analysis to reduce computational complexity.
- For assessing the model through accuracy, precision, recall, F1 score, confusion matrix, and ROC curves.
- For increasing the interpretability of the model using methods of Explainable AI (XAI), such as SHAP and/or LIME.
- For creating a working and understandable system for cybersecurity use.

## 6.Methodology/Proposed System

The proposed system is designed to be a multi-class intrusion detection mechanism based on machine learning. The following stages are covered during the development process, starting from the preparation and collection of datasets and ending with explainable attack classification.

## 6.1 Dataset Collection

NSL-KDD Dataset link: https://www.kaggle.com/datasets/hassan06/nslkdd

The NSL-KDD dataset is used in this project. The NSL-KDD dataset is an enhanced form of the KDD Cup 99 dataset. It has been extensively used in Intrusion Detection studies since it provides labeled data containing attacks along with normal traffic.
There are various types of attacks present in the dataset including:

- Denial-of-Service (DoS)
- Probe
- Remote-to-Local (R2L)
- User-to-Root
- Normal traffic attack
-

## 6.2 Data Preprocessing

Data preprocessing is done before building the machine learning model. The process involves:

- Categorical attribute handling
- Symbolic-to-numeric conversion
- Deletion of duplicates and irrelevant data
- Normalization, if needed

## 6.3 Feature Selection

Feature selection is performed to decrease complexity and increase efficiency. Since there are numerous attributes in the dataset, not all attributes have an equal role in detecting attacks. Selecting features will assist in:

- Quick training
- Improving Efficiency
- Decreasing overfitting
- Interpretation

## 6.4 Model Training

Machine learning models are used to train the system with preprocessed dataset. The algorithms may include:

- Random forest
- Decision tree
- K-nearest neighbors
- Support vector machine
- The model classifies the network traffic into different categories rather than binary classification.

## 6.5 Explainability

To make the model more understandable, interpretability techniques like feature importance are used. It aids in identifying the network attributes that affect predictions the most.

## 6.6 Evaluation

The performance of the machine learning model is measured through:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

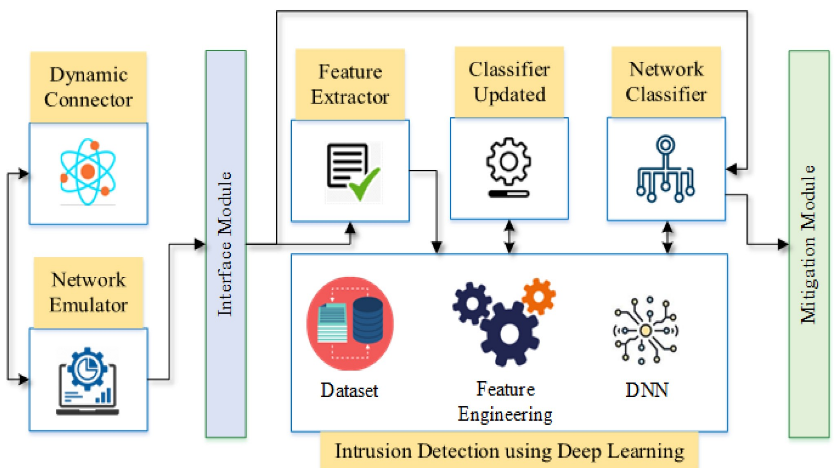


Figure 2: A Novel Deep Learning-Based Intrusion Detection System for IoT Networks

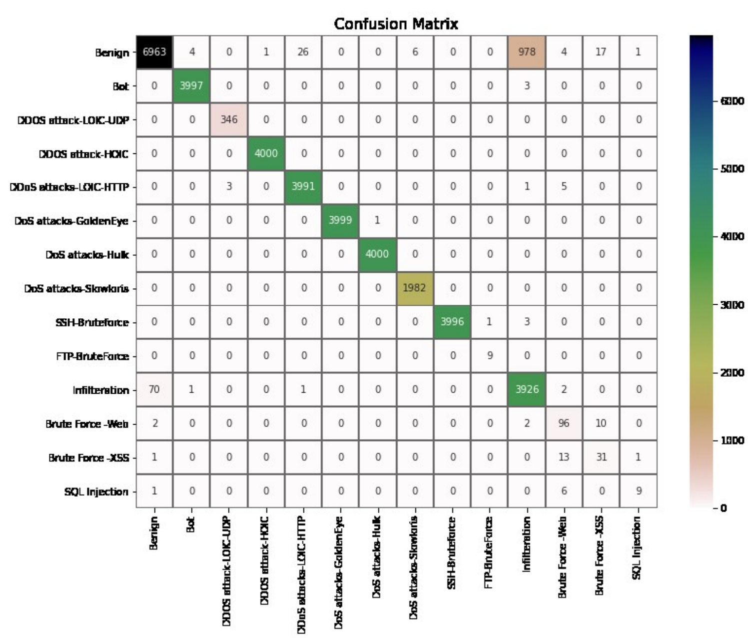


Figure 3: Confusion Matrix of CIC-IDS-2018 by DSSTE+minVGGNet.

# 7. Implementation

The proposed system implementation is performed with the help of Python-based machine learning methods and algorithms. The system is made up of the following modules:

## 7.1 Data Loading Module

NSL-KDD data set is imported into the system with the use of data processing tools, including Pandas and NumPy.

## 7.2 Preprocessing Module

This module includes such tasks as:

- label encoding for categorical variables;
- handling class labeling;
- splitting data into sets for training and testing;
- feature scaling/normalization if needed.

## 7.3 Feature Selection Module

The system uses feature selection methods to find out the most relevant features for intrusion detection.

## 7.4 Classification Module

The machine learning model is trained on the dataset and performs classification of network traffic into categories.

## 7.5 Evaluation Module

Model performance is estimated using confusion matrix and classification metrics after training.

## 7.6 Explainability Module

The explainability module shows contribution of key features in attack identification. This approach helps in understanding why traffic data is labeled to a particular intrusion categories.

**Tools and Technologies Used**

Programming Language: Python

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Dataset: NSL-KDD

Platform: Jupyter Notebook / Google Colab / VS Code

## 9. Results and Analysis

The effectiveness of the proposed intrusion detection system was checked with popular datasets and standard evaluation metrics from machine learning and deep learning. Based on recent studies, hybrid and explainable models appear to be extremely effective at dealing with multi-class attack classification.

A wide variety of models, including CNN, LSTM, Bi-LSTM, Random Forest, XGBoost, and ensemble learning techniques showed good performance at network intrusion detection. However, SHAP and LIME are employed to increase transparency of machine learning models. In recent studies, many models reach impressive results, with 98% accuracy for NSL-KDD dataset, close to 98% for ToN-IoT, around 93% for UNSW-NB15, and more than 98% for X-IIoTID. Some models even exceed 99% accuracy with ensemble learning and specialized setups. It shows that modern intrusion detection systems can be considered very efficient for different environments.

The performance of the intrusion detection system is usually measured with:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- ROC curve
- False positive rate.

Explainability techniques with the help of feature importance makes intrusion detection systems more transparent and reliable. It also allows getting rid of irrelevant features and making a system more efficient.

The findings show that an explainable multi-class intrusion detection system can give impressive performance while also providing concise insights in its decisions which makes it more apt for real-world cybersecurity use cases.

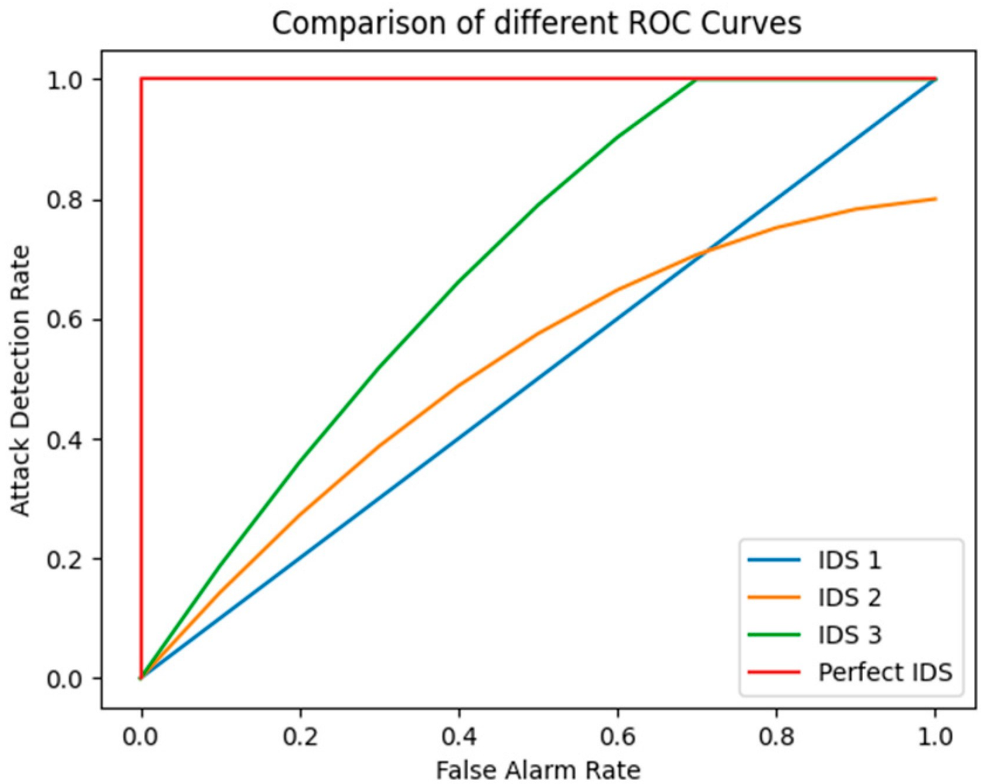


fig 4: ROC Curve Comparison of Different Intrusion Detection Models

## 10. Discussion

From the literature review, one can observe the development of intrusion detection systems, where intelligent ml solutions have replaced classical approaches based on signatures. Advancements in deep learning, ensemble methods, hybrid models, and XAI are among the latest trends making intrusion detection systems increasingly efficient today.

One major trend is moving from binary classification to multi-class classification systems that classify attacks into particular types, rather than just distinguishing normal from malicious traffic.

Another notable trend is the necessity for explainable machine learning. As there are too many high performing black-box models available today, mainly in deep learning, techniques like SHAP and LIME make their decisions more understandable.

Despite all the advancements made, however, the existing methods still face challenges: computationally expensive models, heavy dependency on datasets used, and difficulty in deploying systems in real-time. Also, dealing with imbalanced classes adds to it.

In conclusion, this literature discussion suggests that modern intrusion detection systems must address the issue of explainability, efficiency and being deployable alongside performance.

## 11. Conclusion

This project seeks to design an explainable network intrusion detection system based on multi-class classification algorithms. Research indicates that the performance of ML and DL algorithms as compared to traditional rule-based systems is relatively higher in terms of the ability to detect cyber-attacks.

Most studies agree with the need to use machine learning or deep learning algorithms along with multi-class classification, smart feature selection, hybrid/ensemble model, and the XAI method to build strong systems.

Based on this idea, the proposed model is aimed at providing high performance while remaining interpretable and efficient.

Thus, given the current need for powerful yet transparent cybersecurity tools, an explainable mulitclass intrusion detection system is likely to become an effective tool in protecting against cyber threats.

## 12. Future Scope

This project can be improved and extended in several ways which include:

- Using real-time network traffic data in addition to benchmark datasets
- Applying more advanced deep learning models (CNN, LSTM, etc.)
- Integrating the intrusion detector with live monitoring dashboards
- Improving classification with regard to minority classes like R2L and U2R
- Using the model in cloud/enterprise networks
- Using more advanced methods of explainability (SHAP, LIME)

## 13. Recent References


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